

From Storage Lesion to Biological Signaling: The Emerging Role of Extracellular Vesicles in Transfusion Medicine

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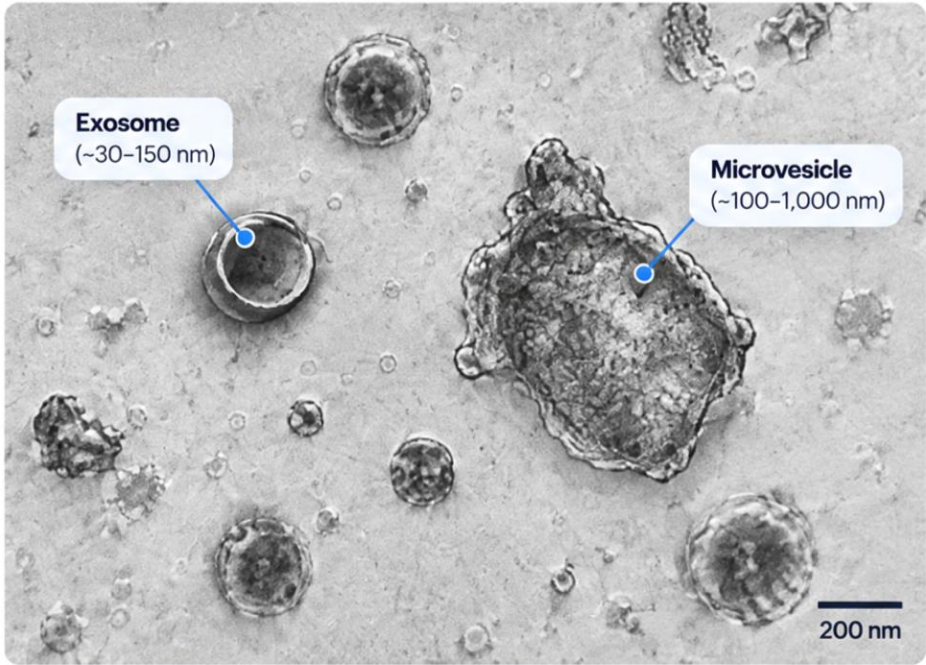
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Introduction: What are Extracellular Vesicles (EVs)?

Extracellular vesicles (EVs) are membrane-bound particles released by almost all cell types and are crucial for intercellular communication. They act as carriers for biologically active molecules.

Contents of EVs:

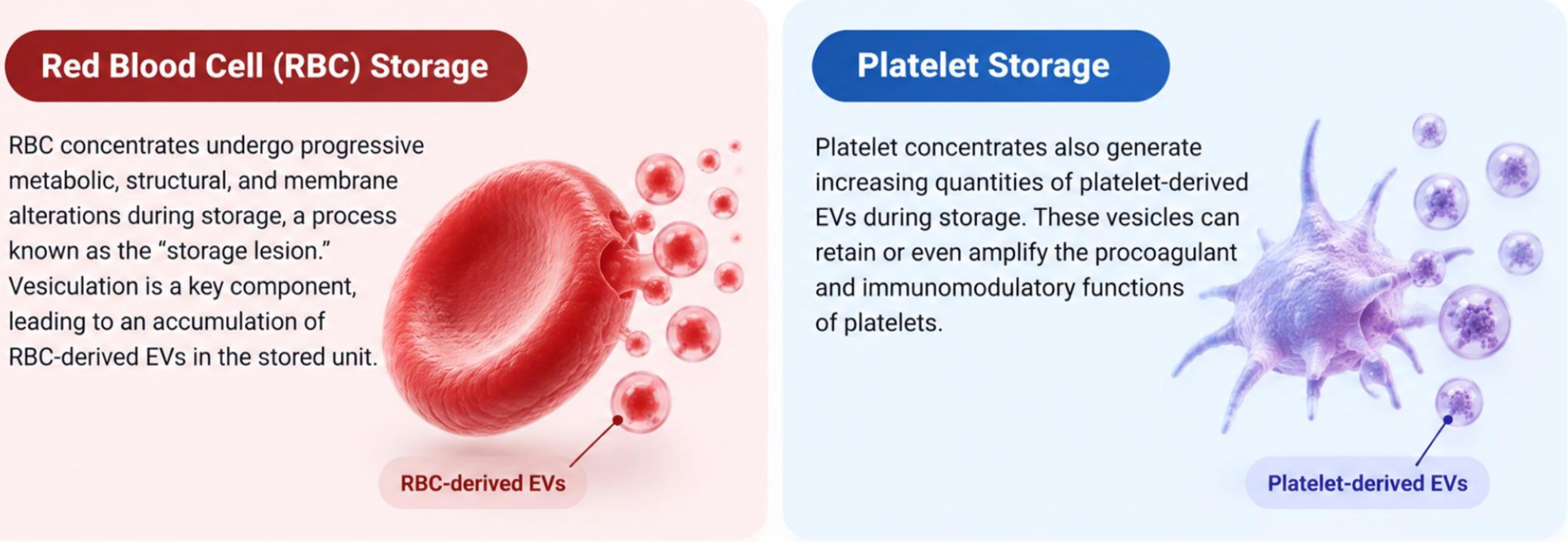


EVs in Transfusion Medicine

In transfusion medicine, EVs are increasingly recognized as important components of stored blood products. During blood collection, processing, and storage, blood cells release EVs in response to various stressors.

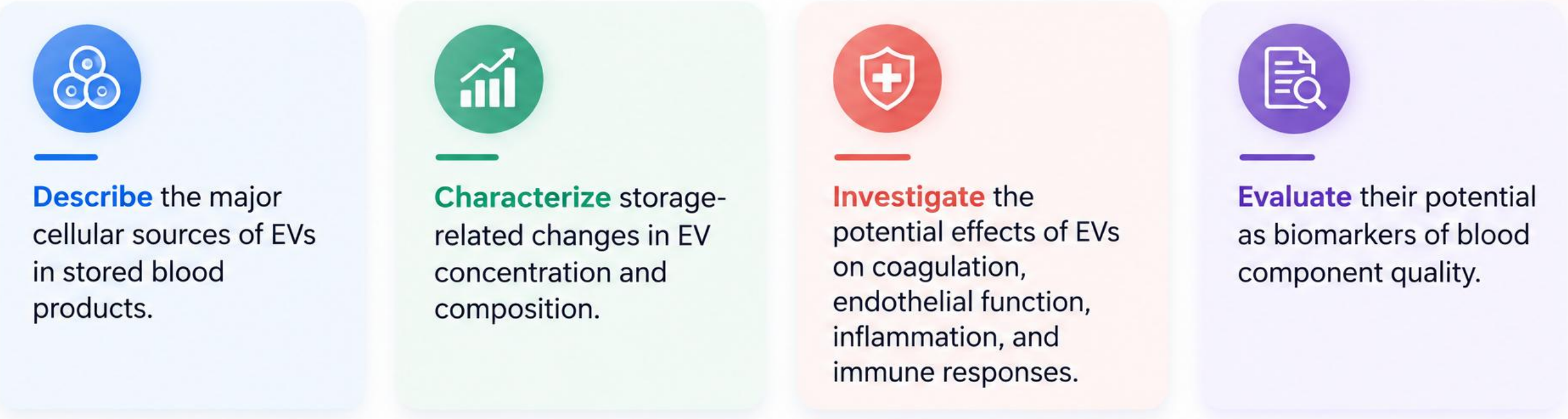
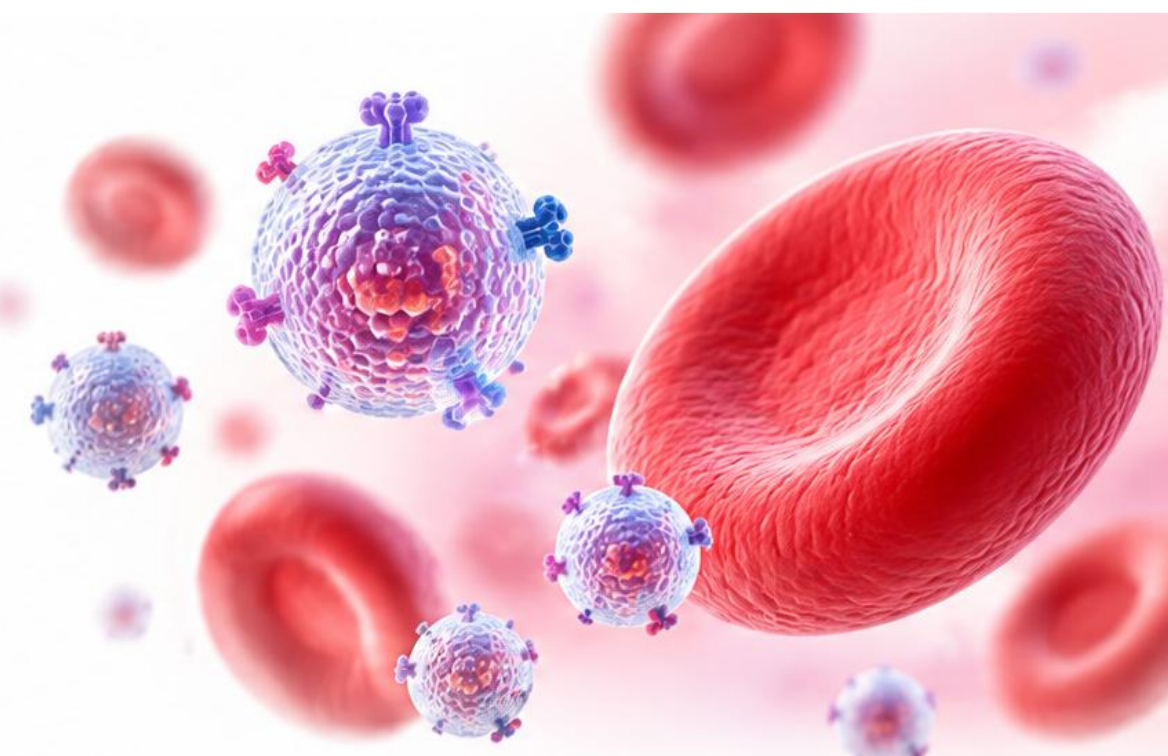


The Storage Lesion and EV Generation



Study Objectives

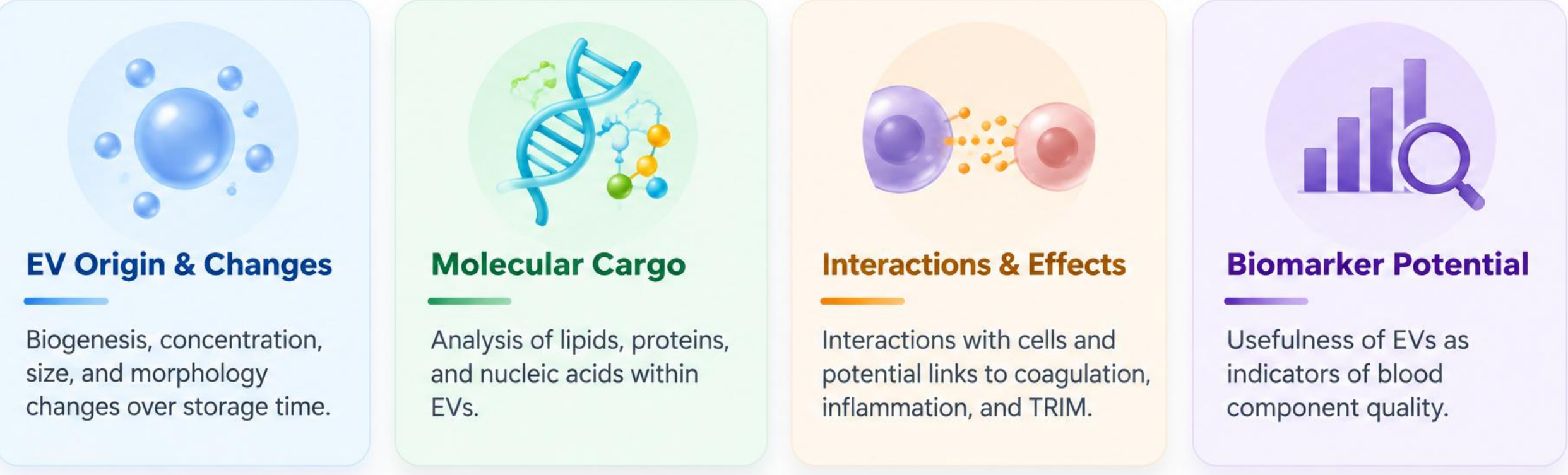
This review aims to synthesize current evidence on the formation, composition, biological functions, and clinical significance of EVs generated during the storage of blood components.



Materials and Methods

A narrative literature review was conducted to synthesize current evidence on extracellular vesicles (EVs) in stored blood components, with a focus on red blood cell (RBC) and platelet concentrates. Recent peer-reviewed studies and relevant reviews were identified through scientific databases.

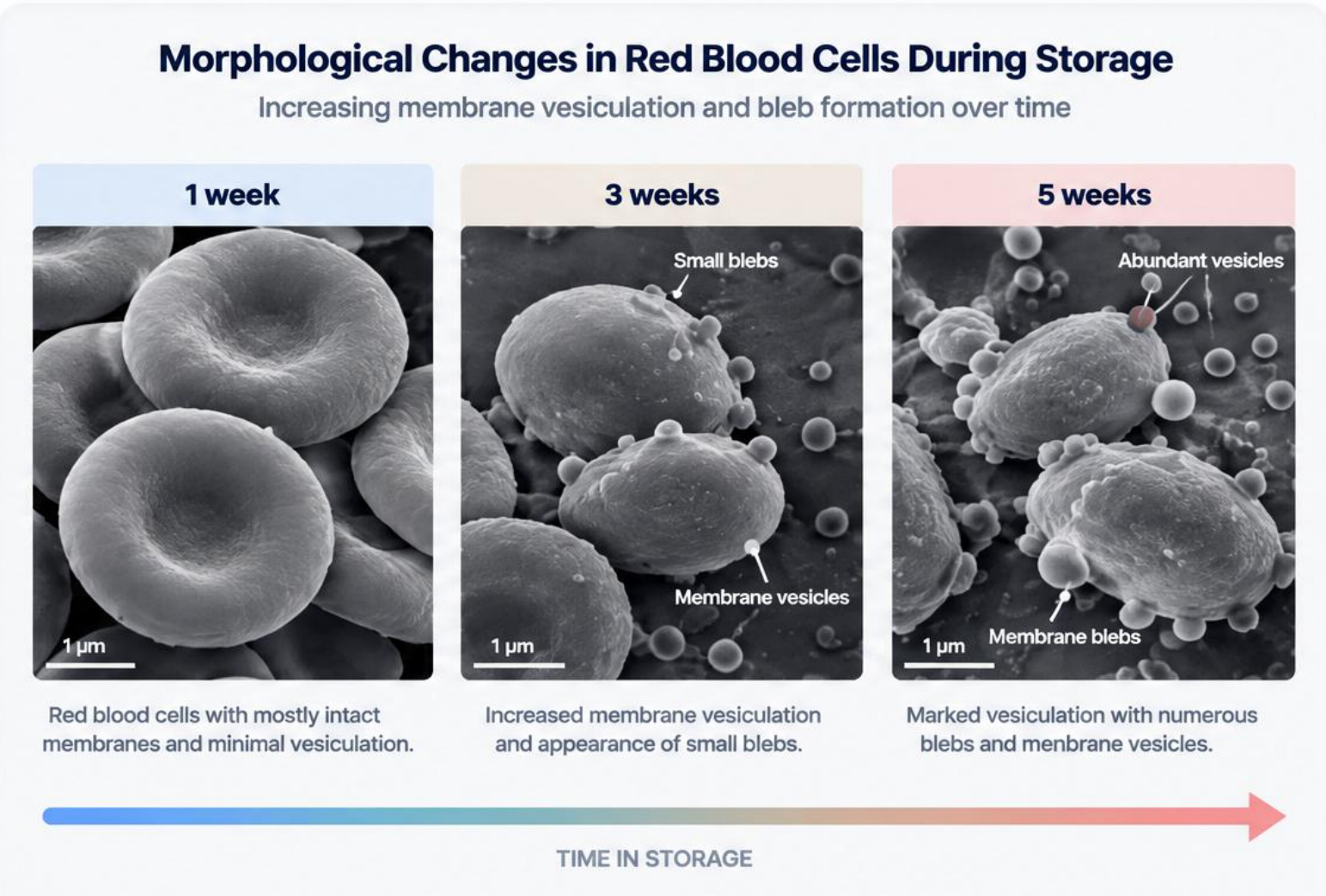
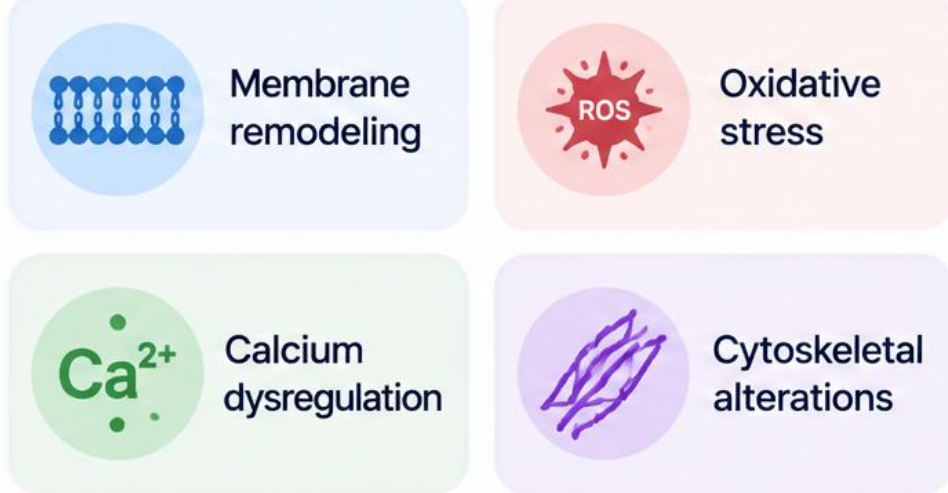
Key Areas of Data Analysis



Results: EV Accumulation During Storage

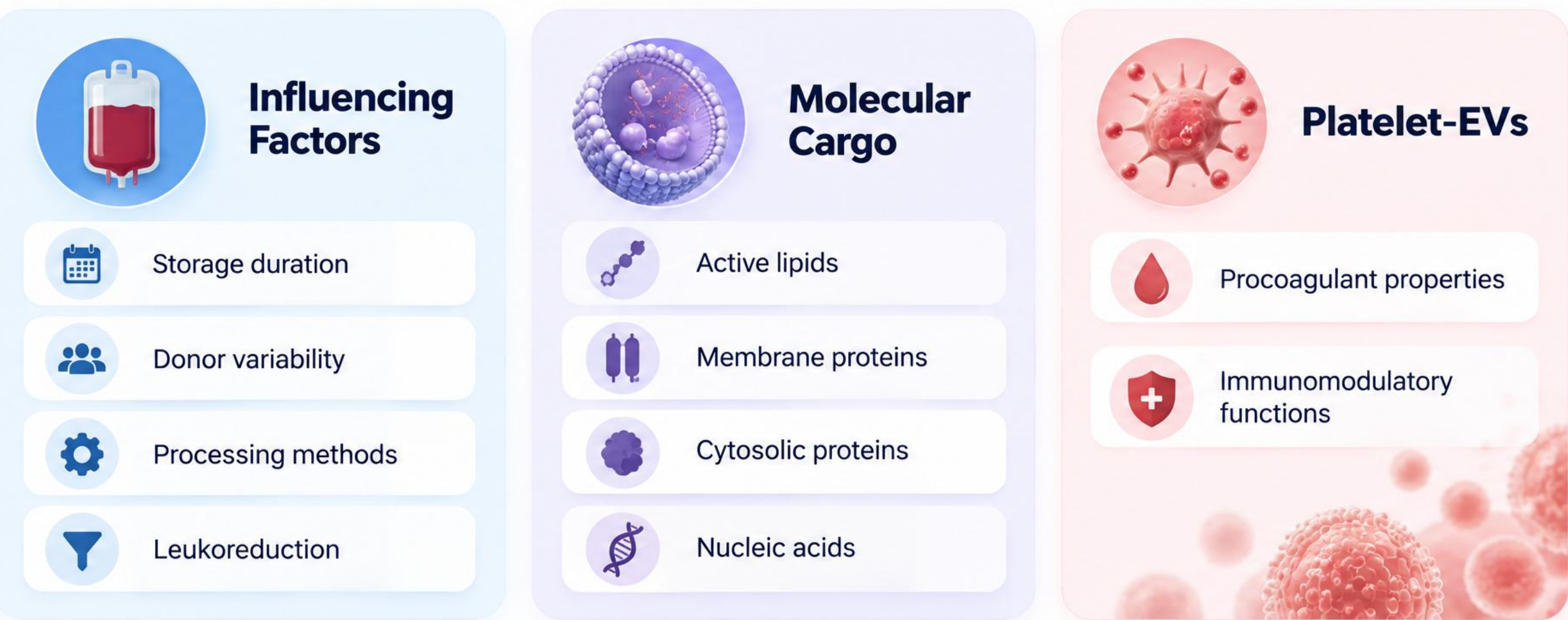
The literature consistently demonstrates that EVs accumulate progressively during the storage of both RBC and platelet concentrates. Their generation is driven by storage-induced cellular stress.

Underlying Causes:



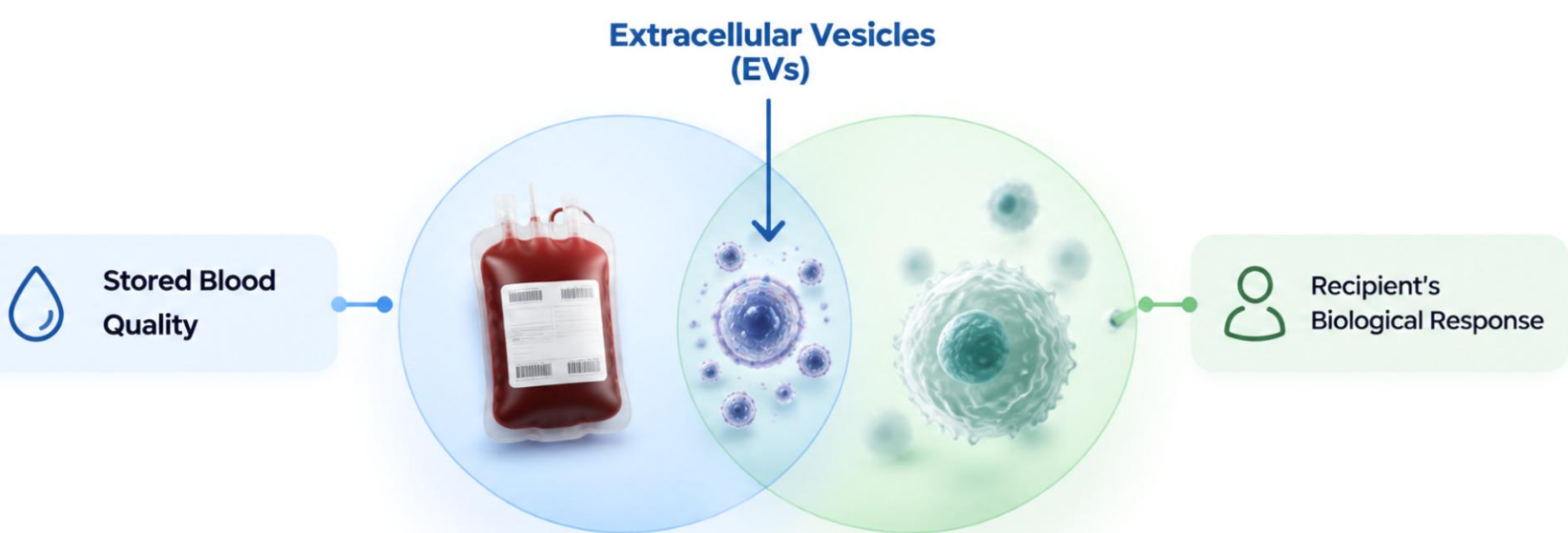
Discussion: Characteristics of Blood-Derived EVs

The concentration and characteristics of EVs change depending on several factors. Their molecular composition reflects the physiological state of the parent cell and the progression of the storage lesion.

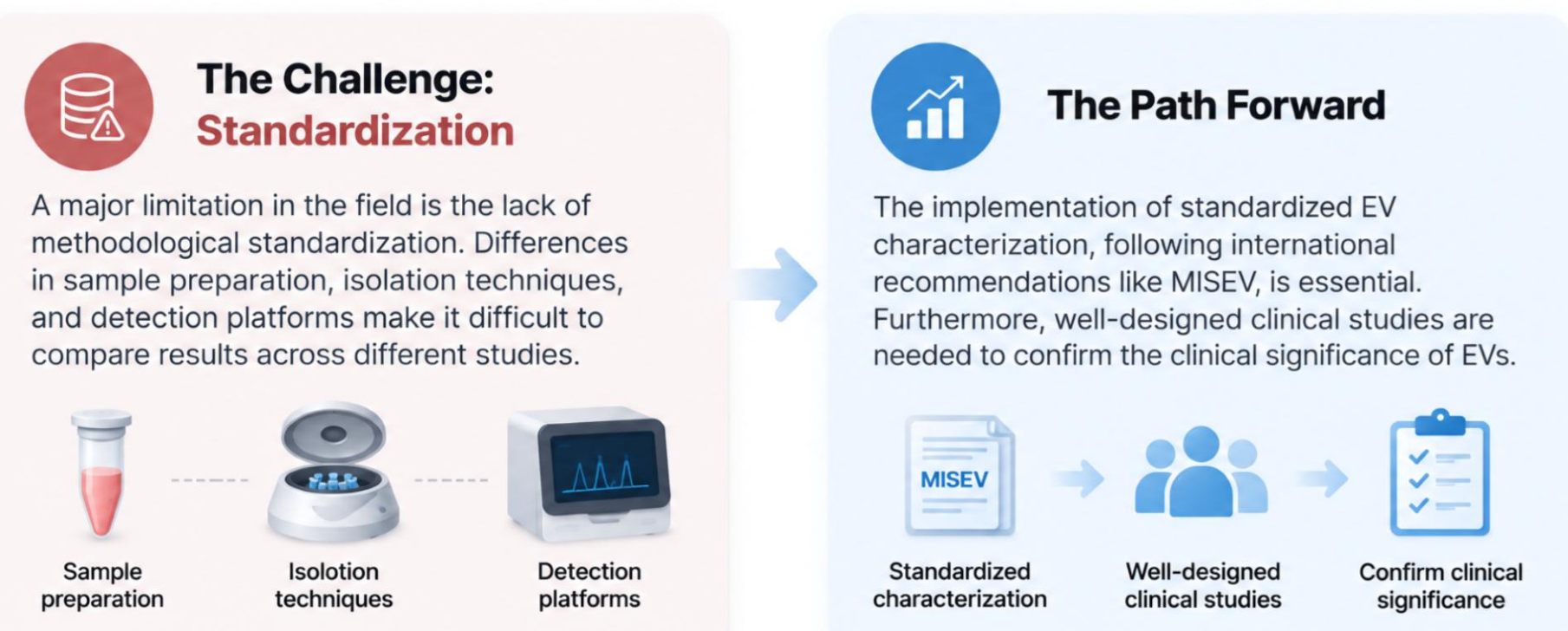


Biological Interactions and a New Paradigm

Experimental studies show that blood-derived EVs interact with various cells, modifying key pathways. This suggests EVs are not passive by-products but active biological mediators linking storage quality to recipient response.

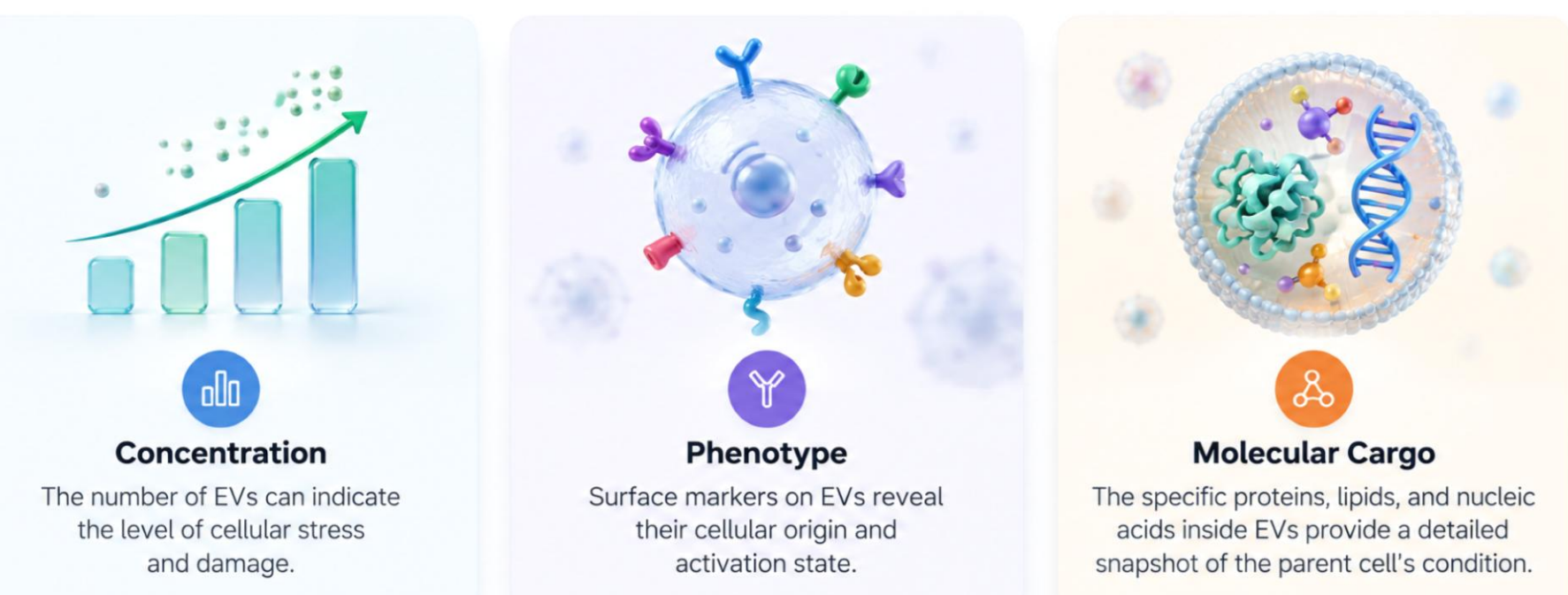


Challenges and Future Directions



Potential Role of EVs as Biomarkers

EVs could offer a more informative assessment of blood component quality than conventional storage parameters. Their characteristics may serve as direct biomarkers of cellular damage and storage-related changes.



Conclusion

- Extracellular vesicles are emerging as biologically active components of stored blood products, not merely passive by-products of cellular degradation.
- Their progressive accumulation and changing molecular composition reflect the evolution of the blood component storage lesion.
- EVs may influence coagulation, endothelial, and immune responses in transfusion recipients.
- Characterizing EVs could provide novel biomarkers for monitoring blood quality and contribute to more precise transfusion strategies.
- Methodological standardization and robust clinical studies are crucial to determine the true clinical significance of EVs and establish them as routine biomarkers.

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